

CSE-2430 Mid Examination Spring 2025

A1

Emergency Evacuation Plan

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Marks : 08

You are the chief organizer of an emergency evacuation in a large city struck by a sudden natural disaster. People in various parts of the city are requesting evacuation, and each request has a priority based on the severity of the situation. Higher-priority requests must be handled before lower-priority ones to save as many lives as possible.

Your task is to manage the evacuation requests using a priority queue. Each time an evacuation team becomes available, you'll handle the highest-priority request. In cases where two requests have the same priority, the one that was submitted earlier should be handled first.

Input Format

- The first line contains an integer q ($1 \leq q \leq 10^5$), the number of requests.
- The next q lines describe each request, which is one of two types:
 - 1 priority - A new evacuation request with an integer priority ($-10^6 \leq \text{priority} \leq 10^6$) is added to the queue.
 - 2 - Process the highest-priority request. Print the priority of this request and remove it from the queue.

Constraints

given on input format

Output Format

For each 2 operation, output the priority of the request being processed.

Sample Output

6
1 5
1 10
2
1 3
2
2

Sample Output

10
5
3

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The Favorite Spellbook

A2

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Marks : 07

In the Great Library of Arcania, mages frequently borrow magical books. Each book has a unique title. The library wants to find out which book has been **borrowed the most times** over the past semester. If multiple books have the highest number of borrows, the library wants to highlight the one that has the largest length.

Input Format:

- The first line contains an integer n ($1 \leq n \leq 10^5$), the number of borrow records.
- The second line contains n space-separated strings, each representing a book title. Book titles contain only lowercase English letters and have a maximum length of 30 characters.

Output Format:

Output a single string: the title of the most borrowed book.

Constraints:

If multiple books have the same highest borrow count, choose the one that is **largest in length**.

Sample Input

```
7
firestorm
shadowbind
firestorm
frostlight
shadowbind
firestorm
shadowbind
```

Sample Output

```
shadowbind
```

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Max Mana Difference

A3

Input: standard input
Output: standard output

Time Limit: 1 second
Memory Limit: 32 MB

Marks : 05

In the realm of Eldoria, each mage maintains a log of their mana levels during a long quest. The levels fluctuate due to battles and resting. You are given an array where each element represents the mana level at a specific point in time.

Your task is to find the **maximum difference** between two mana levels where the **larger value comes after the smaller one**. If the mana level only decreases or stays the same, return 0.

Input Format:

- The first line contains an integer n ($2 \leq n \leq 10^5$), the number of mana recordings.
- The second line contains n space-separated integers, each representing the mana level at a point in time.

Output Format:

Print a single integer — the maximum difference between any two mana levels where the higher value appears **after** the lower value. If no such pair exists, print 0.

Constraints:

- $2 \leq n \leq 10^5$
- $-10^9 \leq \text{mana level} \leq 10^9$

Sample Input 1:

6
3 1 4 8 7 2

Sample Output 1:

7

Explanation:

The maximum difference is $8 - 1 = 7$. 1 comes before 8, satisfying the condition.

Sample Input 2:

5
9 8 7 6 5

Sample Output 2:

0

Explanation:

All mana levels are decreasing, so no valid pair exists where the higher value comes after the lower one.